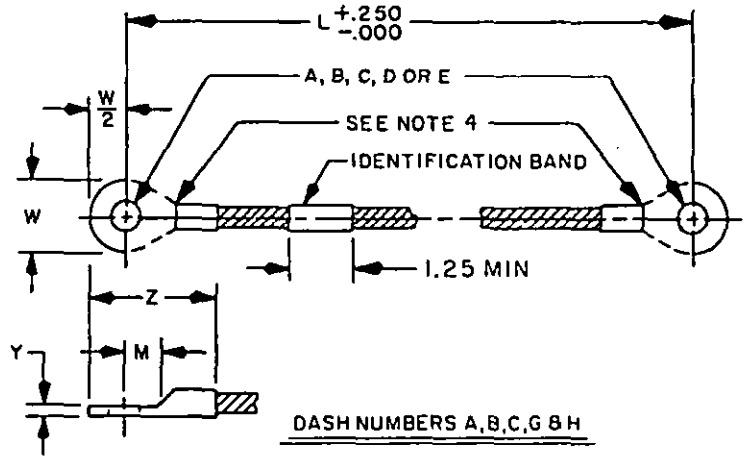
MILITARY SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: TYPE IV JUMPER CABLE ASSEMBLY, LEAD, ELECTRICAL

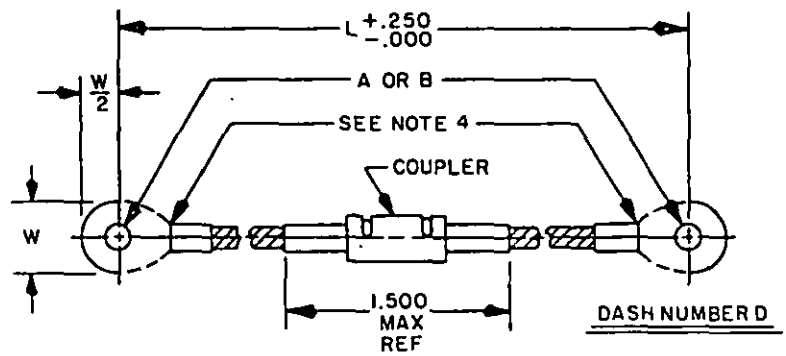
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the assemblies described herein shall consist of this specification and the issue in effect of MIL-C-83413.

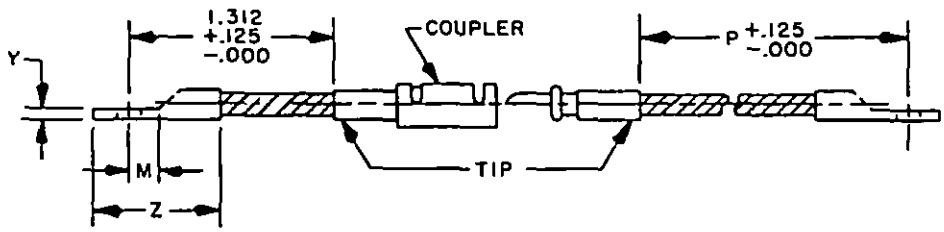


DASH NUMBERS A,B,C,G & H

INCHES	MM
.031	.79
.125	3.18
.250	6.35
.375	9.52
1.500	38.10



DASH NUMBER D



DASH NUMBER E
SHORT END

DASH NUMBER F
LONG END

FIGURE 1. Jumper cable assembly.

FSC 6150

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REQUIREMENTS:

Material:

Copper grounding wire: Per QQ-W-343, type RD of the size specified in table III.

Copper terminals: Per WW-T-799, under the applicable MS sheets specified in table II.

Coupler (1st dash no. "D", "E" and "F" only): Per MIL-S-6852-3.

Identification band: Polyolefin, class 2, heat shrinkable per MIL-I-23053/5.

Color: See table I.

Assembly resistance: See table II.

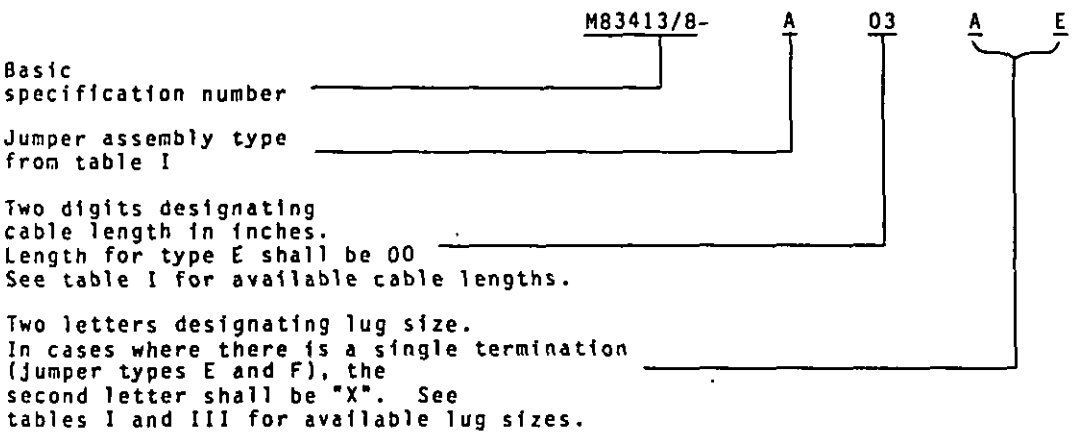
Tensile strength: See table II.

Crimping tools: See table II.

Crimping dies: See table II.

Marking: Part number identification shall be marked on the identification band. 1st dash number "C" parts shall be marked with such an ink or fluid which is compatable with fuel.

Part number: The part number shall consist of the prefix "M83413/8-" and a dash number consisting of four characters, formulated as follows:



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TABLE I. Jumper assembly types and characteristics.

Jumper assembly type	Construction style	Wire and terminal material	Wire size (see table II)	Available sizes (see table III)	Available cable length (±5%, in inches) (see "L" or "P" dimensions on figure 1)	Lightning tested	Fuel 1/ compatible	Method of attaching terminals	Color of ID band
A	Bonding	Copper	12	A, B, C, D, E	02-99	No	No	Crimp	None
B	Current return	Copper	8	D, E	02-99	No	No	Crimp	None
C 2/	Bonding	Aluminum	10	A, B, C, D, E (aluminum only)	04-99	No	Yes	Brazed	Clear
D 3/	Quick disconnect	Copper	12	A, B	03-99	No	No	Crimp	None
E 3/	Short end of quick disconnect	Copper	12	A, B	Fixed length	No	No	Crimp	None
F 3/	Long end of quick disconnect	Copper	12	A, B	02-99	No	No	Crimp	None
G	Bonding	Copper	12	A, B, C, D, E	04-99	Yes	No	Crimp	Yellow
H	Current return	Copper	8	D, E	04-99	Yes	No	Crimp	Yellow

1/ Only fuel compatible jumper assemblies may be used within fuel tank and cells.

2/ Do not use outisde of fuel tanks and cells.

3/ Inactive for Air Force Airborne use after 9 December 1963.

TABLE II. Wire characteristics by jumper assembly type.

Jumper assembly type	Material	Stranding	Wire size	Assembly resistance (max-ohms)	Tensile strength (min-lb)	Copper only	
						Crimping tools	Crimping dies
A, G	Copper	7 X 37	12	.00016XL +.00034	110	MS3316 or MS17776	---
B, H	Copper	7 X 95	8	.00006XL +.00016	225	MS25441	MS23002 MS90485
C	Aluminum	1/	10	.00042XL +.00110	75		
D	Copper	7 X 37	12	.00016XL +.00034	110	MS3316 or MS17776	---
E	Copper	7 X 37	12	.00057	110		---
F	Copper	7 X 37	12	.00016XP +.00034	110		---

1/ Aluminum wire - size 10 (3710.0167), alloy 1350, concentric stranded pitch 3/4 to 1 inch. Dip braze terminals per MIL-B-7883. A coating is not required on interior strands of cable. Chemical conversion coating per MIL-C-5541, class 3, shall be applied after brazing. Invert gas welding may be used with approval of the qualifying activity.

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TABLE III. Available lug sizes.

Code letter lug size designation	For stud size	Aluminum terminals 1/								Copper terminals	
		Stud hole I.D.		M	W		Y		Z		
		Max	Min	Min	Max	Min	Max	Min	Max	MS25036	MS20659
A	.112 or .137	.152 (3.86)	.142 (3.61)	.218 (5.54)	.406 (10.31)	.313 (7.95)	.083 (2.11)	.037 (.94)	1.172 (29.77)	-111	-165
B	.164 or .190	.203 (5.16)	.193 (4.90)	.250 (6.35)						-112	-105
C	.24	.285 (7.24)	.260 (6.60)	.281 (7.14)	.540 (13.72)	.450 (11.93)			1.300 (33.02)	-157	---
D	.3125	.343 (8.71)	.320 (8.13)	.329 (8.36)	.741 (18.82)	.531 (13.49)	.090 (2.29)	.038 (.97)	1.390 (35.31)	2/-113, 3/-117	2/-106, 3/-108
E	.375	.410 (10.41)	.385 (9.78)	.343 (8.71)					1.626 (41.30)	2/-114, 3/-118	2/-128, 3/129

1/ Aluminum terminals shall be of alloy QQ-A-225/2, temper O.
2/ Size #12- #10 AWG wire only.
3/ Size #8 AWG wire only.

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Resistance readings shall be taken at the junction of terminal barrels and tongues.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army - ME, AR, CR, MI
Air Force - 85, 99, 18, 82, 90
DLA - ES, GS

User activity:
Navy - MC

Agent:
DLA - ES

Preparing activity:
Navy - AS

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